



ALFRED UNIVERSITY

SCHOOL OF ENGINEERING AND APPLIED TECHNOLOGY

Department of Applied Electronics and Embedded Technology

Learning Resource Manual

Teaching Media, Study, Reading, and Reference Architecture

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STATUS	Current Governing Document

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Learn. Build. Serve. | Current runtime: v16.3.64 | Core curriculum release: 16.3

CURRENT WEBSITE USE - v16.3.64

This edition replaces the old MUST/SHOULD/STRETCH resource taxonomy. Current resources use Classroom Required, Study, and Engineering Library roles. Alfred remains the primary teacher.

1. What This Manual Controls

This manual explains how to use the external instructional layer without turning AU-ESET 301 into a pile of disconnected courses. The live Teaching Media architecture is the source of truth for assignment role and contextual placement.

Metric	Current baseline
Total retained assignments	347
Classroom Required	247
Study	81
Engineering Library	19
Unique source IDs	322
Canonical sources	319
Target lesson sections	491
Inline placements	1,021
Outside-literature sources in central index	152
Total central reading records	214
Orphaned Required resources	0
Required classification contradictions	0
Required week misalignments	0

2. Three Resource Roles

Role	Purpose	What the learner owes
Classroom Required	A necessary first-pass external explanation, demonstration, written reference, or source for the specific section being taught.	Complete it as part of the Required Classroom path when assigned. Do not treat it as optional just because it is external.
Study	Alternate explanation, repetition, extra example, review, or targeted help after the concept has been reached.	Use when useful. It can be extensive without adding Required workload.
Engineering Library	Professional/deep references: standards, specs, datasheets, manuals, application notes, protocols, toolchain docs, systems engineering, and career references.	Consult when the work requires professional detail. Not every Library item is a course assignment.

3. Dual-Channel Coverage Rule

Every substantive teaching section should have complete first-pass support in both video and outside written literature when those media materially support learning. Cross-medium overlap is desirable. Within a medium, avoid multiple Required resources that teach the same thing unless each makes a distinct necessary contribution.

- One strong video may cover several related sections if it genuinely teaches them.
- One strong written source may cover several related sections if it genuinely supports them.
- Alternate teachers, repeated demonstrations, extra examples, and optional depth belong in Study.
- Authoritative standards/specifications/safety sources can outrank pedagogical convenience when exact wording or boundaries matter.
- Required resources appear centrally and contextually at the exact lesson section where they support learning.

4. Teaching Media Completion Is Not Mastery

NON-NEGOTIABLE

Watched / Read / Reviewed is a shared organizational checklist. It does not complete a lesson, pass a lab, raise an assessment score, or establish competency by itself. Mastery remains evidence-based.

5. CETa Study Guide Integration

Role	Pages / use
Required	64 printed pages across 14 weeks; used only where the book provides a necessary current first-pass contribution.
Study / Review	133 printed pages used for reinforcement and optional review.
Reference / Historical	27 printed pages retained for context where newer official/current material controls.
Errata / authority	Current ETA competencies, formulas, and current official guidance control when the older book conflicts or is incomplete.
Navigation	Printed page -> bundled PDF page uses +9 for the verified copy. Opening the book never counts as mastery/completion.

6. Source Selection Standard

1. Accuracy and currentness first.
2. For beginner teaching, clarity, visuals, pacing, worked examples, and comprehension come before prestige once accuracy is acceptable.
3. Use official/manufacturer/standards sources for exact specifications, safety boundaries, APIs, errata, and device behavior.
4. Avoid dead links, obsolete documentation, unexplained advanced material, and resources that duplicate an existing Required source without adding a distinct contribution.
5. Treat source availability as changeable. The live course can replace a resource while preserving the underlying learning objective and evidence requirement.

7. Domain Resource Strategy

Domain	Resource strategy
Electrical foundations	Beginner electronics teacher + written circuit text; manufacturer/safety references for measurement hazards.
Test equipment	Instrument-maker training plus visually clear demonstrations; measurement planning must precede probe placement.
Semiconductors / analog	Ground-up device models, operating regions, practical measurements, and datasheet use.
Digital / computer architecture	Number systems, logic, state, timing, memory, and interfaces taught before MCU dependencies.
C / embedded C	General C first; then memory, bitwise reasoning, interrupts/events, and hardware-facing patterns.
STM32 / debug	Official STM32 toolchain documentation plus beginner-friendly demonstrations for bring-up, SWD, GPIO, timers, UART, and buses.
Python test automation	pySerial / test framework / plotting docs used to produce repeatable hardware evidence, not generic software exercises.
Professional practice	Standards, datasheets, workmanship references, systems engineering, documentation, Git, and role-facing test/reporting evidence.

8. How to Use Study Without Creating a Backlog

- Use the Study week selector to review any reached week without changing Classroom progress.
- When you are stuck, choose one alternate explanation or one extra worked example before opening a long list of resources.
- Use Active Recall and changed-context practice to test whether the resource actually transferred into usable knowledge.
- Stop collecting sources once you can explain, predict, calculate, measure, build, debug, or document the target skill.

9. Resource Governance

Current Teaching Media architecture is controlled by the v16.3.60 combined multimodal QA repair. The centralized reading index, Study Guide mapping, contextual placement, and source-role metadata are supporting records. Historical media acceptance reports remain historical evidence and must not be presented as the current whole-system verification.